Differentiation
MTH 201 - Business Calculus

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Outline

1. Derivative of functions
2. Differentiation techniques
3. The chain rule
4. Higher-order derivatives

Average rates of change

If a car travels 110 mi in 2 hr, its *average rate of change (speed)* is 110 mi/2 hr, or 55 mi/hr (55 mph). Suppose you accelerate on a freeway and, glancing at the speedometer, you see that at that instant your *instantaneous rate of change* is 65 mph. These are two quite different concepts. The first you are probably familiar with. The second involves ideas of limits and calculus.

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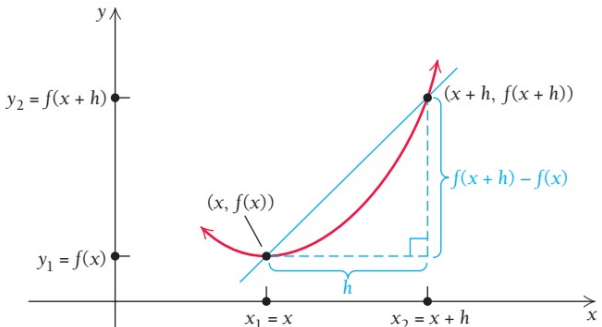
Definition

The average rate of change of y with respect to x, as x changes from x_1 to x_2, is the ratio of the change in output to the change in input:

$$\frac{y_2 - y_1}{x_2 - x_1}, \quad \text{where } x_1 \neq x_2.$$

Difference quotient as average rates of change

Let $y = f(x)$. we write $x_1 = x$ and $x_2 = x + h$

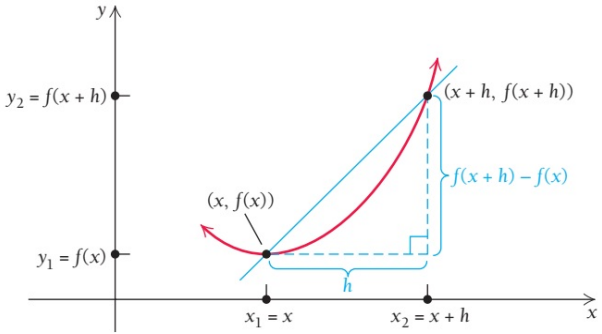


$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{f(x_2) - f(x_1)}{x_2 - x_1} = \frac{f(x + h) - f(x)}{x + h - x} = \frac{f(x + h) - f(x)}{h}$$ difference quotient.

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Example 1. Find the simplified difference quotient for each function listed.

a) $f(x) = x^2$ b) $f(x) = \frac{1}{x^2}$ c) $f(x) = \sqrt{x}$.

Difference quotient as average rates of change

Example 2. For $f(x) = x^3$, find a simplified form of the difference quotient.

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Solution.

$$\frac{f(x+h) - f(x)}{h} = 3x^2 + 3xh + h^2, \quad h \neq 0.$$

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When $x = 2$ and $h = 0.1$,

$$\frac{f(x+h) - f(x)}{h} = 12.61.$$

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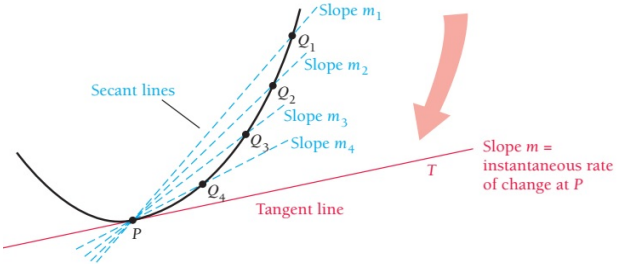
Tangent lines

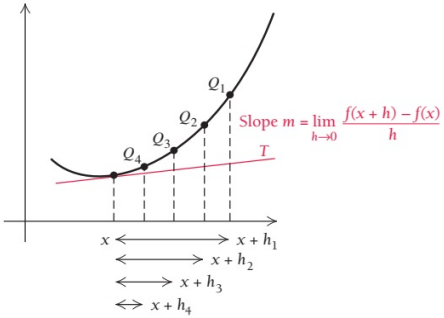
Consider secant lines through P and neighboring points Q_1, Q_2, Q_3, Q_4, and so on. As the Q's approach P, the secant lines approach line T.

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Tangent lines

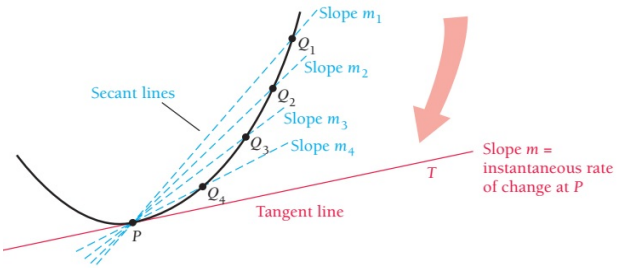
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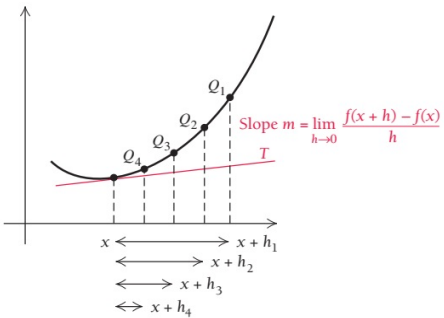




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The slope of the tangent line at $(x, f(x))$ is $m = \lim_{h\to 0} \dfrac{f(x+h)-f(x)}{h}$.

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Derivative

Definition

For a function $y = f(x)$, its derivative at x is the function f' defined by

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h},$$

provided that the limit exists. If $f'(x)$ exists, then we say that f is differentiable at x. We sometimes call f' the derived function.

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The notation $f'(x)$ is read "the derivative of f at x," "f prime at x," or "f prime of x."

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Example. Find the derivative of $f(x) = x^3$, $g(x) = \sqrt{x}$, $p(x) = \dfrac{1}{x}$, $q(x) = \ln x$.

Exercise 1. Let f be a piecewise-defined function given by

$$f(x) = \begin{cases} x^3, & \text{for } x \leq 1, \\ 3x - 2, & \text{for } x > 1. \end{cases}$$

a) Verify that f is continuous at $x = 1$.

b) Is f differentiable at $x = 1$? Explain why or why not.

Exercise 2. Let g be a function given by

$$g(x) = \begin{cases} 2x^2 - x, & \text{for } x \leq 3, \\ mx + b, & \text{for } x > 3. \end{cases}$$

Determine the values of m and b that make $g(x)$ differentiable at $x = 3$?

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Derivative

Leibniz notation

If $y = f(x)$, then the derivative of y with respect to x is $\dfrac{dy}{dx} = f'(x)$.

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Historical Note: *The German mathematician and philosopher Gottfried Wilhelm von Leibniz (1646–1716) and the English mathematician, philosopher, and physicist Sir Isaac Newton (1642–1727) are both credited with the invention of calculus, though each worked independently. Newton used the dot notation $\dot{y}$ for dy/dt, where y is a function of time; this notation is still used, though it is not as common as Leibniz notation.*

Leibniz - Newton



Gottfried Wilhelm von Leibniz (1646 - 1716)



Sir Isaac Newton (1642 - 1727)

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The power rule

Theorem

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Differentiation techniques

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The derivative of a constant function is 0. That is, $\dfrac{d}{dx}c = 0$.

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Theorem

The derivative of a constant times a function is the constant times the derivative of the function

$$\frac{d}{dx}[c\,f(x)] = c\frac{d}{dx}f(x).$$

Theorem (The sum-difference rule)

Sum. *The derivative of a sum is the sum of the derivatives*

$$\frac{d}{dx}[f(x)+g(x)] = \frac{d}{dx}f(x) + \frac{d}{dx}g(x).$$

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Differentiation techniques

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Example 1. Differentiate $f(x) = 2x^7 - 3\sqrt[5]{x} + \dfrac{1}{2x^3} + 2.$

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Example 1. Differentiate $f(x) = 2x^7 - 3\sqrt[5]{x} + \dfrac{1}{2x^3} + 2.$

Example 2. Use the derivative to help explain why $f(x) = x^5 + x^3$ increases for all x in $(-\infty, \infty)$.

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Exercise. (Growth of a baby). The median weight of a boy whose age is between 0 and 36 months is approximated by

$$w(t) = 8.15 + 1.82t - 0.0596t^2 + 0.000758t^3,$$

where t is in months and w is in pounds.

(Source: Centers for Disease Control. Developed by the National Center for Health Statistics in collaboration with the National Center for Chronic Disease Prevention and Health Promotion, 2000, Rechecked 2014)

Use this approximation to find the following:

a) The rate of change of weight with respect to time.

b) The weight of a boy at age 10 months.

c) The rate of change of a boy's weight with respect to time at age 10 months.

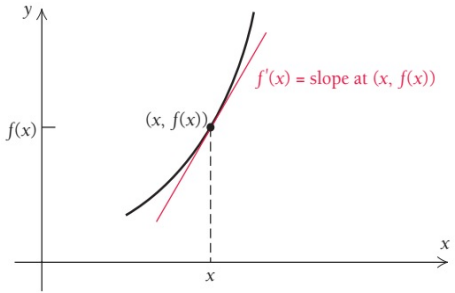
Tangent lines

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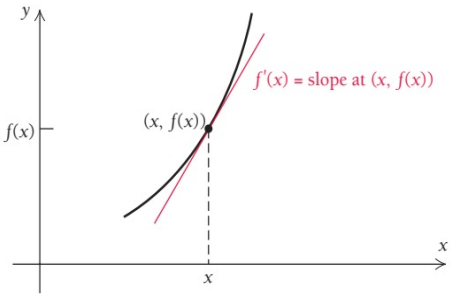
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Example 1. Find the points on the graph of $f(x) = -x^3 + 6x^2$ at which

a) the tangent line is horizontal. b) the tangent line has slope 9.

Example 2. Find the equation of the tangent line to the graph of $f(x) = \sqrt[3]{x^2} + x$ at $(-1; 0)$.

Example 3. Find the equation of the tangent line to the graph of $g(x) = \dfrac{8}{\sqrt{x-2}}$ at $(6; 4)$.

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Differentiation techniques

Analyzing a function by its derivative

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Example. Use the derivative to help explain why $f(x) = x^3 + ax$ increases for all x in $(-\infty, \infty)$ when $a \geq 0$ but not when $a < 0$.

Differentiation techniques

The product rule

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The product rule

Theorem

Let $F(x) = f(x).g(x)$. *Then*

$$F'(x) = \frac{d}{dx}[f(x).g(x)] = f(x)\left[\frac{d}{dx}g(x)\right] + g(x)\left[\frac{d}{dx}f(x)\right].$$

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Theorem

If $Q(x) = \dfrac{N(x)}{D(x)}$, *then*

$$Q'(x) = \frac{D(x)N'(x) - N(x)D'(x)}{[D(x)]^2}.$$

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$N(x)$ numerator, $D(x)$ denominator.

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Differentiation techniques

Example. Differentiate

a) $f(x) = (x^2 + 3x)\ln x$

b) $g(x) = x\sqrt{4 - x^2}$

c) $h(x) = \dfrac{x^3 + 2}{x^2 - 1}$

d) $p(x) = \dfrac{x}{\sqrt{x^2 + 1}}$

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If $C(x)$ is the cost of producing x items, then the average cost of producing x items is $\dfrac{C(x)}{x}$. If $R(x)$ is the revenue from the sale of x items, then the average revenue from selling x items is $\dfrac{R(x)}{x}$.

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If $P(x)$ is the profit from the sale of x items, then the average profit from selling x items is $\dfrac{P(x)}{x}$.

Example (Exercise 60, Textbook). Cruzin' Boards has found that the cost, in dollars, of producing x skateboards is given by $C(x) = 900 + 18x^{0.7}$. If the revenue from the sale of x skateboards is given by $R(x) = 75x^{0.8}$, find the rate at which the average profit per skateboard is changing when 20 skateboards have been built and sold.

Differentiation techniques

The power rule

The power rule

Theorem (The extended power rule)

Suppose that $g(x)$ is a differentiable function of x. Then, for any real number k,

$$\frac{d}{dx}[g(x)]^k = k[g(x)]^{k-1}\frac{d}{dx}g(x).$$

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Example. Differentiate

$$f(x) = (x^3 + x + 1)^5, \quad g(x) = \sqrt[3]{x^2 + x + 1}, \quad h(x) = \left(\frac{x^2}{8} + x - \frac{1}{x}\right)^4, \quad r(x) = \sqrt[4]{\frac{x+1}{x-2}}.$$

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The chain rule

Theorem (The chain rule)

The derivative of the composition $f \circ g$ is given by

$$\frac{d}{dx}\left[(f \circ g)(x)\right] = \frac{d}{dx}\left[f(g(x))\right] = f'(g(x)) \cdot g'(x).$$

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The chain rule often appears in another form: For $y = f(u)$ and $u = g(x)$ then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}.$$

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Example 1.

$$\begin{pmatrix}\text{Change in profits}\\ \text{with respect to time}\end{pmatrix} = \begin{pmatrix}\text{Change in profits with respect}\\ \text{to number of games produced}\end{pmatrix} \cdot \begin{pmatrix}\text{Change in number of games}\\ \text{produced with respect to time}\end{pmatrix}.$$

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Example 1. Game Boss Video's profit, in dollars, is given by $y = f(u) = 0.1u^2 - 500$, where u is the number of units sold. Its sales are given by $u = g(x) = 125x + 40$, where x is the number of days. Find the rate at which profit is changing on the 5th day.

Answer.

Theorem (The chain rule)

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Answer. We have $y = (f \circ g)(x)$... For $x = 5$, we have $\frac{dy}{dx} = \$16625$ per day.

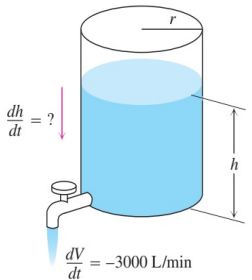
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The chain rule

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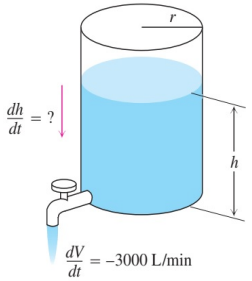


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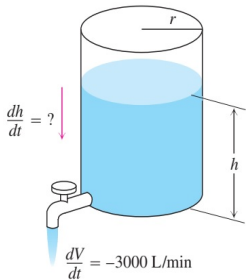


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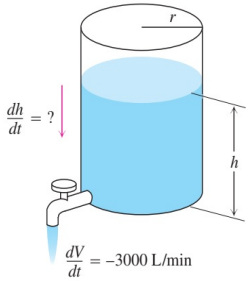
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Answer. Let h and V be the height and the volume of the fluid respectively. We have $V = 1000\pi r^2 h$ (because a cubic meter contains 1000L)... The fluid level will drop at the rate of $3/(\pi r^2)$ m/min.

The chain rule

Exercise 1. (Cost, revenue, and profit) A company can manufacture x items at a cost of $c(x)$ thousand dollars, a sales revenue of $r(x)$ thousand dollars, and a profit of $p(x) = r(x) - c(x)$ thousand dollars. Find dc/dt, dr/dt, and dp/dt for the following values of x and dx/dt.

a) $r(x) = 9x$, $c(x) = x^3 - 6x^2 + 15x$, and $dx/dt = 0.1$ when $x = 2$

b) $r(x) = 70x$, $c(x) = x^3 - 6x^2 + 45/x$, and $dx/dt = 0.05$ when $x = 1.5$.

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Exercise 2. (Changing dimensions in a rectangle) The length ℓ of a rectangle is decreasing at the rate of 2 cm/sec while the width w is increasing at the rate of 2 cm/sec. When $\ell = 12$ cm and $w = 5$ cm, find the rates of change of $(\mathbf{a})$ the area, $(\mathbf{b})$ the perimeter, and $(\mathbf{c})$ the lengths of the diagonals of the rectangle. Which of these quantities are decreasing, and which are increasing?

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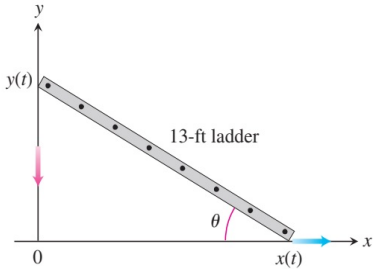
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Exercise 3. (The radius of an inflating balloon) A spherical balloon is inflated with helium at the rate of 100π ft^3/min. How fast is the balloon's radius increasing at the instant the radius is 5 ft? How fast is the surface area increasing?

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The chain rule

Exercise 4. (A sliding ladder) A 13-ft ladder is leaning against a house when its base starts to slide away. By the time the base is 12 ft from the house, the base is moving at the rate of 5 ft/sec.



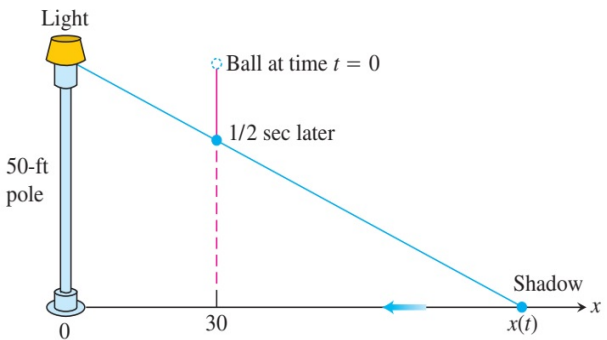
a) How fast is the top of the ladder sliding down the wall then?

b) At what rate is the area of the triangle formed by the ladder, wall, and ground changing then?

c) At what rate is the angle θ between the ladder and the ground changing then?

The chain rule

Exercise 5. (A moving shadow) A light shines from the top of a pole 50 ft high. A ball is dropped from the same height from a point 30 ft away from the light. (See accompanying figure.) How fast is the shadow of the ball moving along the ground $1/2$ sec later? (Assume the ball falls a distance $s = 16t^2$ ft in t sec.)



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The chain rule

Exercise 6. For

$$y = \frac{u-1}{u+1} \quad \text{and} \quad u = \sqrt[3]{x}.$$

Find dy/du, du/dx, and dy/dx.

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$$f(u) = 2\sqrt[3]{\frac{u}{u^2+1}} \quad \text{and} \quad g(x) = u = \frac{x}{x+1}.$$

Find $(f \circ g)'(x)$.

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Exercises. Page 164: 97-109, page 170: 23-26 (see textbook).

Outline

1. Derivative of functions
2. Differentiation techniques
3. The chain rule
4. **Higher-order derivatives**

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Higher-order derivatives

Consider the function $y = f(x) = x^5 + 3x^3 - 2x$.

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For $f(x) = x^5 + 3x^3 - 2x$, then

$$y'' = f''(x) = 20x^3 + 18x.$$

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Higher-order derivatives

Continuing in this manner, we have

$$
\begin{aligned}
f'''(x) &= 60x^2 + 18, && \textit{The third derivative of } f,\\
f''''(x) &= 120x, && \textit{The fourth derivative of } f,\\
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Note $f^{(n)}(x) = \dfrac{d^n}{dx^n} f(x)$.

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Exercise 4. If $f(x) = \dfrac{x^{50} + x^{49} + 2}{x+1}$, evaluate $f^{(50)}(2)$. Express your answer using factorial notation.

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Velocity and acceleration

Definition

The velocity of an object that is $s(t)$ units from a starting point at time t is given by

$$\text{Velocity} = v(t) = s'(t) = \lim_{h \to 0} \frac{s(t+h) - s(t)}{h}.$$

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where $s(t)$ is in meters (m).
The velocity of the object is given by $v(t) = s'(t) = 9.81t$.

Velocity, acceleration and jerk

Definition

Acceleration is the derivative of velocity with respect to time. If a body's position at time t is $s = f(t)$, then the body's acceleration at time t is

$$a(t) = \frac{dv}{dt} = \frac{d^2s}{dt^2}.$$

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Jerk *is the derivative of acceleration with respect to time:*

$$j(t) = \frac{da}{dt} = \frac{d^3s}{dt^3}.$$

Velocity, acceleration and jerk

Example 1. Galileo's experiments with free fall

$$s(t) = \frac{1}{2}gt^2 = 4.905t^2,$$

where g is the acceleration due to Earth's gravity:

$$g = \frac{d^2}{dt^2}s(t) = 9.81m/sec^2(\text{ meters per second squared}).$$

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The jerk of gravity is zero.

Higher-order derivatives

Example 2. Assume that the function $N(t) = \dfrac{250000t^2}{(2t+1)^2}$, $t > 0$, represented the quantity sold N of a product after t weeks on the market. Its derivative is

$$N'(t) = \frac{500000t}{(2t+1)^3}.$$

$N'(t)$ represented the *rate of change* in number of units sold per week. Find $N''(t)$.

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